Ziwei Gu

CONTACT INFORMATION

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RESEARCH INTERESTS

Human-Computer Interaction (HCI)
Natural Language Processing, Machine Learning, Human-AI Interaction

EDUCATION

Harvard University, Cambridge, Massachusetts

Ph.D. Computer Science, *August 2022 – Present*
Advised by Elena Glassman, Assistant Professor of Computer Science

Cornell University, Ithaca, New York

B.A. Computer Science, **Magna cum laude**, *August 2017 – December 2020*
B.A. Mathematics, *August 2017 – December 2020*

PEER-REVIEWED PAPERS

Ziwei Gu, Joyce Zhou, Ning-Er (Nina) Lei, Jonathan K. Kummerfeld, Mahmood Jasim, Narges Mahyar, and Elena L. Glassman. AbstractExplorer: Leveraging Structure-Mapping Theory to Enhance Comparative Close Reading at Scale. *In ACM Symposium on User Interface Software and Technology (UIST)*, 2025.

Ziwei Gu, Ian Arawjo, Kenneth Li, Jonathan K. Kummerfeld, and Elena L. Glassman. An AI-Resilient Text Rendering Technique for Reading and Skimming Documents. *In ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2024.

Katy Ilonka Gero, Chelse Swoopes, **Ziwei Gu**, Jonathan K. Kummerfeld, and Elena L. Glassman. Supporting Sensemaking of Large Language Model Outputs at Scale. *In ACM CHI Conference on Human Factors in Computing Systems (CHI),* 2024. Honorable Mention Award (top 5%)

Ziwei Gu, Owen Raymond, Naser Al Madi, and Elena L. Glassman. Why Do Skimmers Perform Better with Grammar-Preserving Text Saliency Modulation (GP-TSM)? Evidence from an Eye Tracking Study. *In ACM CHI Conference on Human Factors in Computing Systems Late Breaking Work (CHI LBW),* 2024.

Ziwei Gu*, Gauri Jain*, Hongwen Song*, Isak Diaz*, Margaux Masson-Forsythe*, and Jorge Valdes. BiomeAzuero2022: A Fine-Grained Dataset and Baselines for Tree Species Classification with Ground Images. *In the 37th AAAI Conference on Artificial Intelligence (AAAI-23) AI for Social Good Workshop*, 2023.

Jing Nathan Yan, **Ziwei Gu**, and Jeffrey M Rzeszotarski. Tessera: Discretizing Data Analysis Workflows on a Task Level. *In ACM CHI Conference on Human Factors in Computing Systems (CHI),* 2021.

Ziwei Gu*, Jing Nathan Yan*, and Jeffrey M Rzeszotarski. Understanding User Sensemaking in Machine Learning Fairness Assessment Systems. *In WWW'21: The Web Conference (WWW)*, 2021.

Jing Nathan Yan, **Ziwei Gu**, Hubert Lin, and Jeffrey M Rzeszotarski. Silva: Interactively Assessing Machine Learning Fairness Using Causality. *In ACM CHI Conference on Human Factors in Computing Systems (CHI),* 2020.

*: Equal contribution

Research Experience

Harvard University, Boston, Massachusetts — *2022-Present*

- Developed and evaluated an LLM-powered recursive sentence compression algorithm for grammar-preserving extractive summarization and text saliency modulation visualization, which led to significantly more efficient reading comprehension compared to existing methods.
- Created and evaluated an LLM-powered interactive system for analyzing research abstracts at scale (>1,000); validated through ablation and summative studies and demonstrated significant improvements in reading and skimming efficiency on a large research corpus.

Cornell University, Ithaca, New York — *2019-2020*

- Adapted the Transformer model for open information extraction, modeled as a sequence-to-sequence transduction task. Our model was competitive with state-of-the-art systems but did not rely on other NLP tools.

Industry Experience

Human Frontier Collective Fellow, **Scale AI** — *2025-Present*

- Designed complex, domain-specific problems and datasets to evaluate and enhance the accuracy, reasoning, and overall performance of advanced LLMs.

Data Scientist Intern, **Lyft**, San Francisco, California — *2020*

- Analyzed millions of rider payment transactions; applied clustering to identify user profiles and proposed a Lyft Family feature, which was implemented and remains active in the app.
- Designed a Markov chain model to study driver churn and estimated transition probabilities between active and churn states; quantified the statistical significance of incentive effects and delivered insights that informed driver engagement strategies.

Teaching Experience

Head Teaching Fellow, COMPSCI 178 Engineering Usable Interactive Systems, Harvard CS — *2023*
Graduate Teaching Assistant, CS 4410 Operating Systems, Cornell CS — *2021*
Teaching Assistant, CS 4780 Machine Learning, Cornell CS — *2019-2020*
Teaching Assistant, CS 3410 Computer System Organization and Programming, Cornell CS — *2020*
Teaching Assistant, CS 2110 Object-Oriented Programming and Data Structures, Cornell CS — *2018-2019*

Professional Services

Conference Reviewer
CHI (2024, 2025, 2026), UIST (2024, 2025), VIS 2025, Creativity & Cognition 2025

Student Volunteer
IUI 2023

Other Experience

Treasurer, Harvard Chinese Students and Scholars Association (HCSSA) — *2022-Present*
Fellow (Intellectual and Cultural), The Student Center at Harvard Griffin GSAS — *2023-2025*
Project Lead, Statistics Faculty Award winner, Cornell Data Science Team — *2018-2020*
Resident Advisor, Clara Dickson Hall, Cornell University — *2019-2021*

Latest update: 12/2025